

Exercises Modelling

A) UML Class Diagram

Create a UML Class Diagram for following given description:

Meeting Minutes 1: Mr. Wayne (university's teaching representative)

As one of the central aspects of our ULS, students shall be able to communicate with other students, or even with one or more lecturers. Furthermore, lecturers shall also be able to communicate with other lecturers and students. It is necessary to know the course of study, the student ID, and the first and last name of each student who is working with the ULS. Moreover, we need the information regarding the date of appointment, the employee ID, as well as the first and last name of each lecturer. This information is important to be able to authenticate each student and lecturer.

A student can take an exam, but we have to ensure that an exam can only be taken if there are at least 10 students registered for the exam. In addition, the capacities of the available rooms only allow a maximum of 50 students to take an exam at once. Students have to register for exams; they can be distinguished by the date and the duration of the exam. Each exam which is taken by student is graded by a lecturer, and the grades are obviously very important for the students. In addition, the date of exam registration has to be saved due to accounting purposes.

A lecturer gives one or more courses, and each lecture is given by one single lecturer. The course has a corresponding ID and name. Furthermore, the timeslot and the number of credit points have also to be stored. Some courses are only available during summer or winter term, therefore it is necessary to save in which term the course is usually taught. Additionally, some courses are not taught by the same lecturer in every term. Hence, it is important to capture in which term a lecture or an exercise is given by which lecturer.

A student is allowed to register to courses, but does not necessarily have to do so. According to the exam registration mentioned above, it is also necessary to save the date of the registration for each course separately due to traceability issues. We have to ensure that every student who is registered to a course is able to attend to the lecture. This is the reason why we have to limit the course registration numbers to 50 students. Furthermore, a lecture is only allowed to take place if at minimum 10 students are registered to it.

In addition, a course can be accompanied by more than one exam, but there are also courses without exam. Of course, each exam has exactly one corresponding course. In contrast to the realization of a lecture, the grading process can be realized by more than one lecturer.

B) Cardinality Constraints

Detect defects in following UML class diagrams regarding the cardinality constraints:

Diagram 1

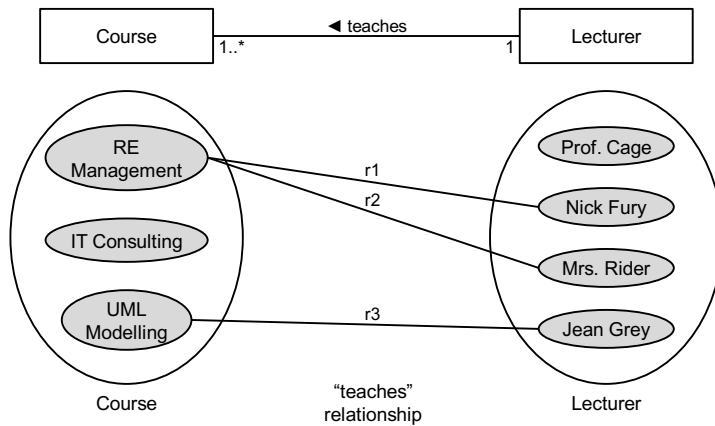


Diagram 2

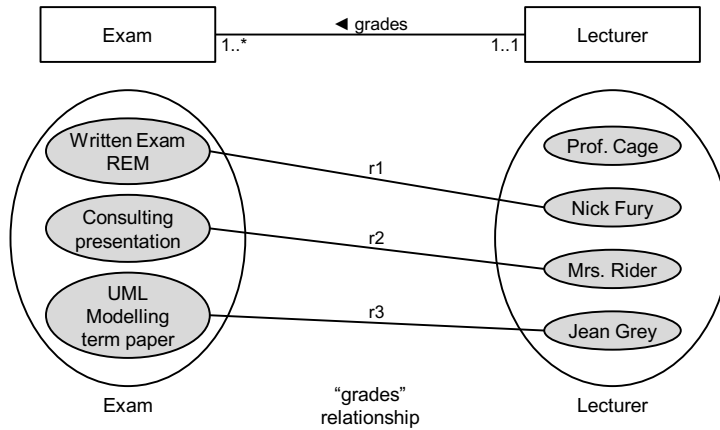
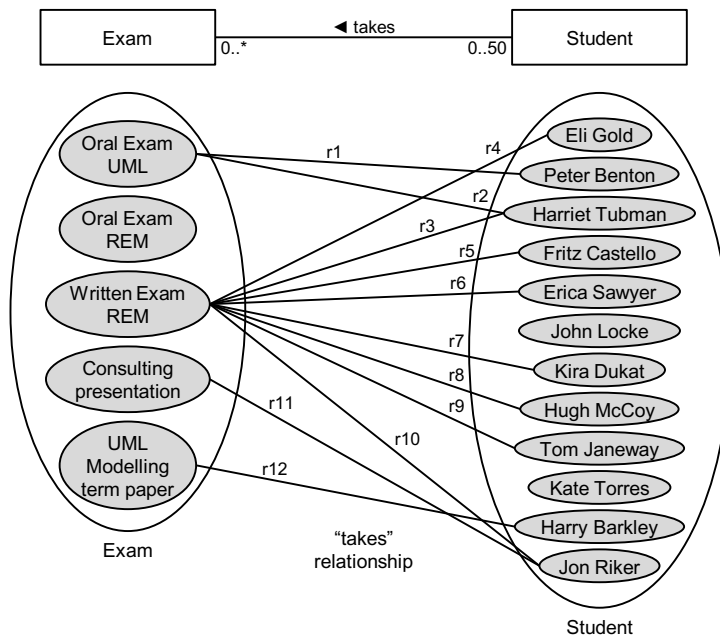


Diagram 3



C) Data Flow Diagram 1

Create a context diagram describing following situation:

The new ULS shall enable the student to register to desired courses and exams, furthermore to review desired results. Moreover, the examination office transfers the permission status of students regarding a specific course and also the course information to the ULS. In contrast to this, the ULS instantly broadcasts the results, student IDs, course IDs, and exam dates to the examination office. These data have been processed in the ULS, as you know from the previous exercises. Finally, the lecturer can enter the examination results in the ULS to store this information in the system.

D) Data Flow Diagram 2

Create a data flow diagram describing following situation (ensure consistency with the context diagram):

The new ULS shall enable the student to view a list of his/her examination results. These results are entered by the lecturer. This way, the student can select a desired exam and the system displays the information. Furthermore, the entered results have to be stored; the examination office is able to access this information.

Moreover, the student can register for desired courses. The courses are stored in the course register. All the necessary course information, e.g., course id, exam information, exam date, or course task are available in the register. The course ID is composed of a four letters of the course of study and a sequence of seven numbers.

The register is provided by the examination office, so it is ensured that the data is always up to date. If a student registers for a course, the course ID and dates have to be saved in the schedule of the student. All dates have got the same structure: it is constructed out of the year, the month, and finally the day. In addition, the dates of a course have to have the currency information. The structure of a year is four numbers; the structure of a month is two numbers, where the first number is zero or one; and the structure of a day is three numbers, where the first number is between zero and three.

A student is able to register to desired exams, hence it is necessary to know the courses the student is registered to. It is therefore necessary to know the student's ID that is a unique sequence of 12 numbers. In addition, the examination office stores the information about the permissions of the student. As soon as a student wants to register to an exam, it has to be ensured that the student fulfils the permissions to participate in the exam. The information about the exam registration has to be saved in the schedule of the student so the student can review the date and all relevant information regarding the exam. Furthermore, the information of the registration has to be transferred to the examination office so it can store the information for traceability reasons. The necessary information for the examination office is the student ID, course ID, and the exam date.

E) UML State Machine Diagram 1

Create a UML State Machine Diagram for an airport self-check-in counter:

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| R1: | The system shall start in a standby mode waiting for user interaction while displaying advertisements. |
| R2: | As welcome screen, the system shall display the main menu when the customer touches the display. The main menu shall provide options to check-in via booking number and last name, using the frequent flyer card or other ID cards. |
| R2.1: | If the customer selects the option to check-in via booking number and last name, the system shall display a form where the booking number and the last name can be typed. |
| R2.1.1: | If the customer fills out the form for his/her booking number and last name, the system shall search for flights while displaying a waiting screen. |
| R2.2: | If the customer inserts his/her frequent flyer card to check-in, the system shall display the frequent flyer menu. |
| R2.2.1: | If the customer selects the option "flights", the system shall search for flights while displaying a waiting screen. |
| R2.3: | If the customer inserts other ID cards, the system shall search for flights while displaying a waiting screen. |
| R3.1: | If the search result is null, the system shall display that no flight was found. |
| R3.1.1: | If no flight was found, the system shall wait for five seconds. |
| R3.1.2: | After five seconds in waiting the system shall display the main menu with the options to check-in via booking number and last name, using the frequent flyer card or other ID cards. |
| R3.2: | If the search result is equal to or higher than one, the system shall display the search result list. |
| R4: | If the customer selects a flight from the search result list, the system shall display the selected flight information. |
| R4.1: | If the seat selection option is available, the customer is able to select the option and the system shall display the menu for selecting a seat. |
| R4.1.1: | If the customer selects a seat, the system shall display the flight information. |
| R4.2: | If the option for eating menu selection is available, the customer is able to select it and the system shall display the menu for meal selection. |
| R4.2.1: | If the customer selects a meal, the system shall display the flight information. |
| R4.3: | If the baggage number is equal to or higher than one, the customer is able to select the option to drop off the baggage. |
| R4.3.1: | If the customer selects to drop off the baggage the system shall display the menu for drop off the baggage. |
| R4.3.2: | If the customer is selecting to print the baggage tag, the system shall display the flight information. |
| R4.4: | If the system displays the flight information, the customer is able to select to print the boarding pass. |
| R4.4.1: | If the customer selects to print the boarding pass, the system shall print the boarding pass and display a printing screen. |
| R4.4.2: | If the boarding pass is printed, the customer can take the boarding pass. |
| R4.4.3: | If the customer takes the boarding pass, the system shall go into standby mode. |

R5:	In case more than one flight has been found. The system shall provide the option to go back to the search results while displaying flight information.
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Consider also following constraints:

R-C7:	The system shall have two modes: inactive and active.
R-C8:	The system shall start in inactive mode.
R-C9:	If the system is booted, the system shall become active.
R-C10:	If the system is active, a shutdown will result in inactiveness.
R-C11:	If the system shows parts of the flight menu, i.e., displays flight information, or the user is selecting her seat, meal, baggage, or boarding pass, the user can cancel his activity and the system will switch to displaying the welcome screen.

F) UML State Machine Diagram 2

Create a UML State Machine Diagram for an Exterior Lighting Control:

R1:	If the driver unlocks the car, the light control component shall turn the ambient light on.
R2:	If the driver closes the DC door, the light control component shall turn off.
R3:	If the engine is running, the driver is able to turn on the light control component and the low beam shall shine.
R4:	If the driver turns off the low beam, the light control component turns off.
R5:	If the low beam is turned on, the driver is able to trigger the high beam switch so the light control component shall turn on the high beam.
R6:	If the high beam is shining and the driver triggers the high beam switch, the low beam is active.
R7:	While the car is turning a corner and the low or high beam is on, the corner light is active.
R8:	While the car is driving straight, the corner light is inactive.
R9:	If the driver pulls the pitman arm up, the right indicator is activated.
R10:	If the pitman arm is in the neutral position and the right indicator blinked three times, the indicator component shall be inactive.
R11:	If the driver pulls the pitman arm down, the left indicator is activated.
R12:	If the pitman arm is in the neutral position and the left indicator blinked three times, the indicator component is inactive.
R13:	If the driver triggers the warning light button, the warning light is activated.
R14:	If the driver triggers the warning light button when warning lights are flashing, the warning light is inactivated.